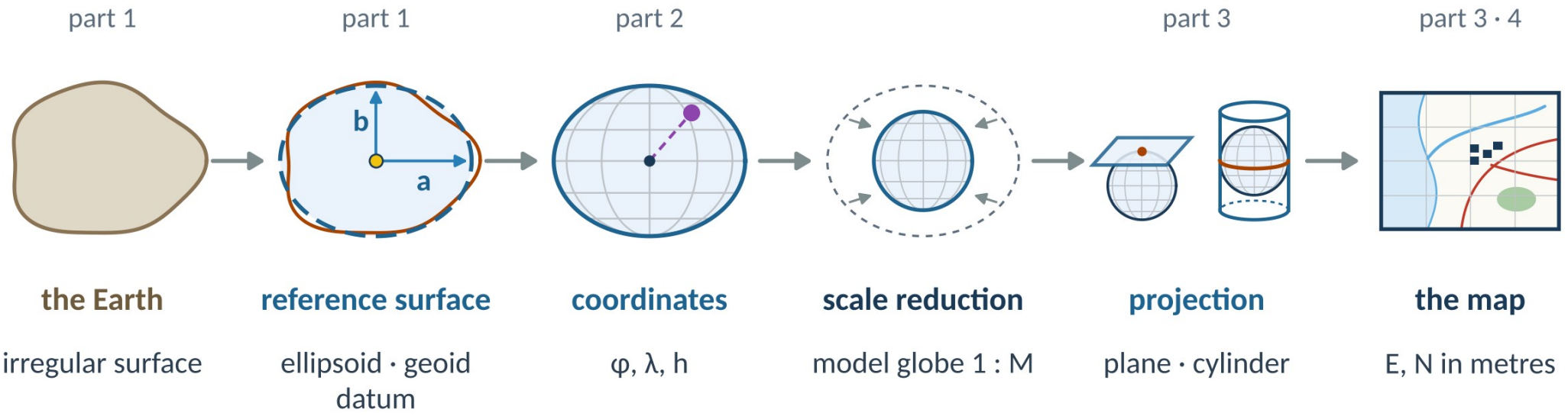
Coordinate reference systems

Shape of the Earth, coordinates, map projections and CRS in practice

Examples: NRW, Kamp-Lintfort



From the Earth to a map



- A coordinate is only meaningful together with its CRS.
- CRS = datum (ellipsoid + position) + coordinate system (+ projection, units, axes)

Shape of the Earth → **Coordinates** → **Map projections** → **Germany / NRW** → **CRS in practice**

Wikipedia: Spatial reference system · Map projection

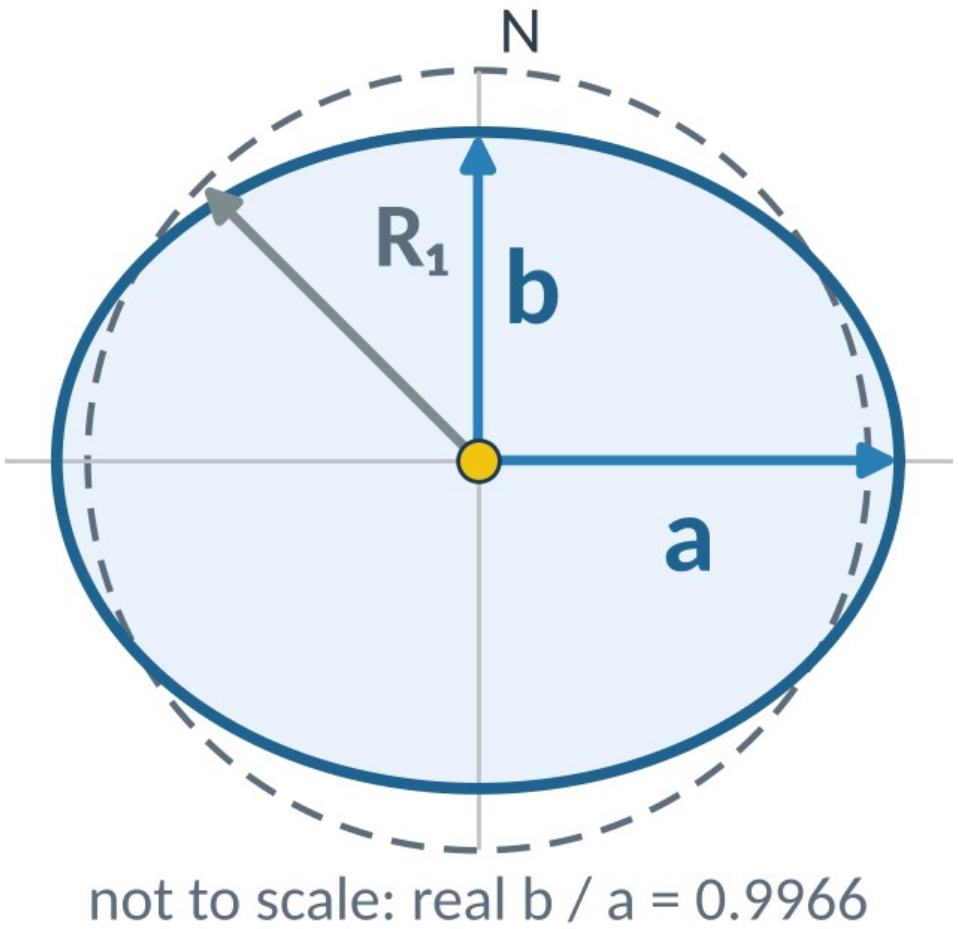
1 Shape of the Earth

ellipsoid · geoid · datum

Shape of the Earth | Coordinates | Map projections | Germany / NRW | CRS in practice



WGS 84: the ellipsoid with a, b and f



- Used by GPS, EPSG:4326
- Origin: Earth's centre of mass
- **a** = 6 378 137 m (equator)
- **b** = 6 356 752 m (pole)
- **f** = (a − b) / a = 1 / 298.257
- R_1 = (2a + b) / 3 = 6 371 km (mean radius)

Wikipedia: World Geodetic System · Flattening

Reference ellipsoids

Ellipsoid	a [m]	b [m]	1 / f	used by (datum · EPSG)
GRS80	6 378 137.000	6 356 752.314	298.257222101	ETRS89 · EPSG:4258
WGS84	6 378 137.000	6 356 752.314	298.257223563	WGS 84 · EPSG:4326 (GPS)
Bessel 1841	6 377 397.155	6 356 078.963	299.1528128	DHDN · EPSG:4314 (DE, old)
Clarke 1866	6 378 206.400	6 356 583.800	294.978698214	NAD27 · EPSG:4267 (US, old)

```
from pyproj import CRS, Geod
CRS("EPSG:25832").ellipsoid      # GRS 1980
g = Geod(ellps="GRS80"); g.a, g.b, 1/g.f
```

a – b ≈ 21.4 km

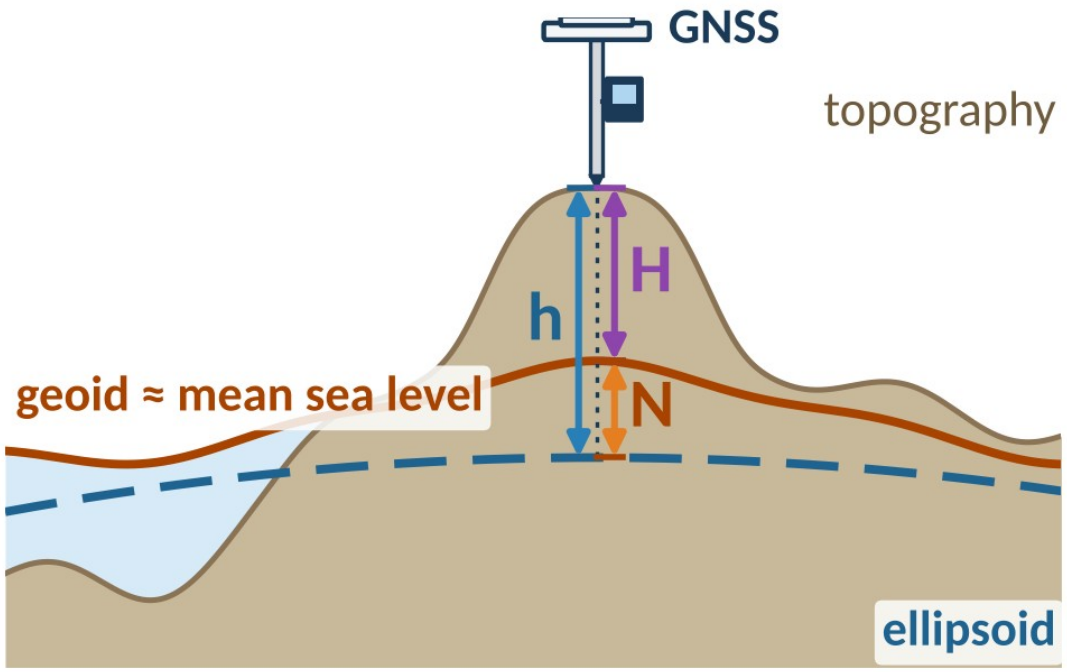
flattening f = (a – b) / a

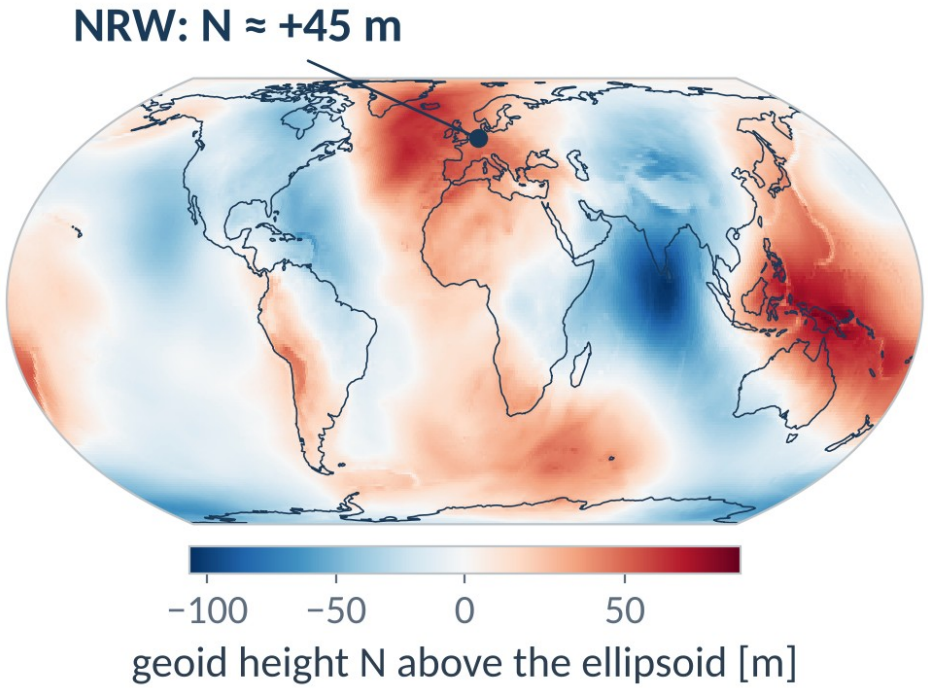
f ≈ 1 : 298, round to 0.3 %

Wikipedia: Earth ellipsoid · GRS 80 · World Geodetic System · Flattening



The geoid: where is 'zero height'?





$$h = H + N$$

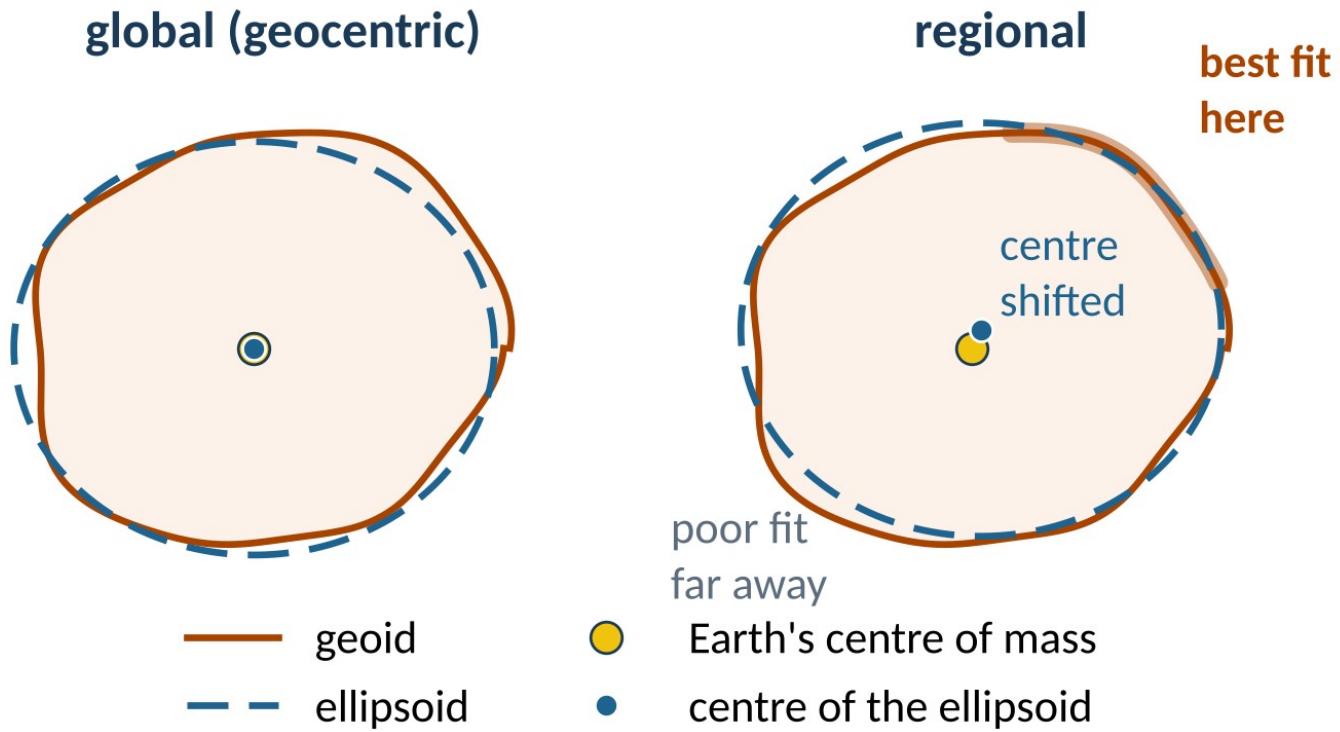
h GNSS **H** maps, DEM **N** geoid height

NRW: GNSS height ≈ map height + 45 m

Wikipedia: Geoid · Earth Gravitational Model · Orthometric height

Geoid: EGM96 (NGA/NASA, public domain) via PROJ-data. Coastlines and borders: Natural Earth (public domain). Cross-section schematic, vertical scale exaggerated.

Datum: where does the ellipsoid sit?



- **Global:** centre = centre of mass, used by GNSS
- **Regional:** fits one country closely (Bessel / DHDN), centre ≈ 0.7 km off

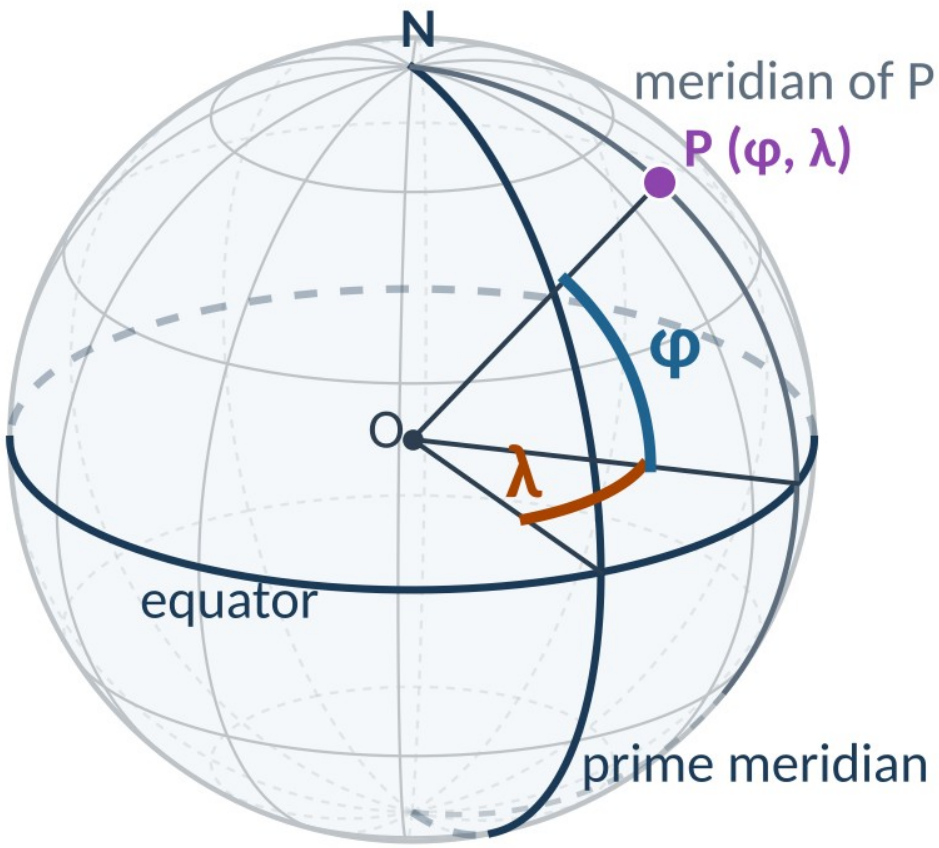
Wikipedia: Geodetic datum · Figure of the Earth

2 Coordinates

φ, λ, h · X, Y, Z



Geographic coordinates: latitude φ and longitude λ



- **λ longitude:** east + / west −, from the prime meridian, −180° … +180°
- **φ latitude:** north + / south −, from the equator, −90° … +90°
- **h** ellipsoidal height in m (part 1)
- **Kamp-Lintfort:** (φ, λ) = (51.5°, +6.53°) = (51.5° N, 6.53° E)

On the ellipsoid φ is the angle of the surface normal, not of the line to the centre (difference up to 0.19°).

Wikipedia: Geographic coordinate system · Latitude · Longitude · Prime meridian

Writing degrees

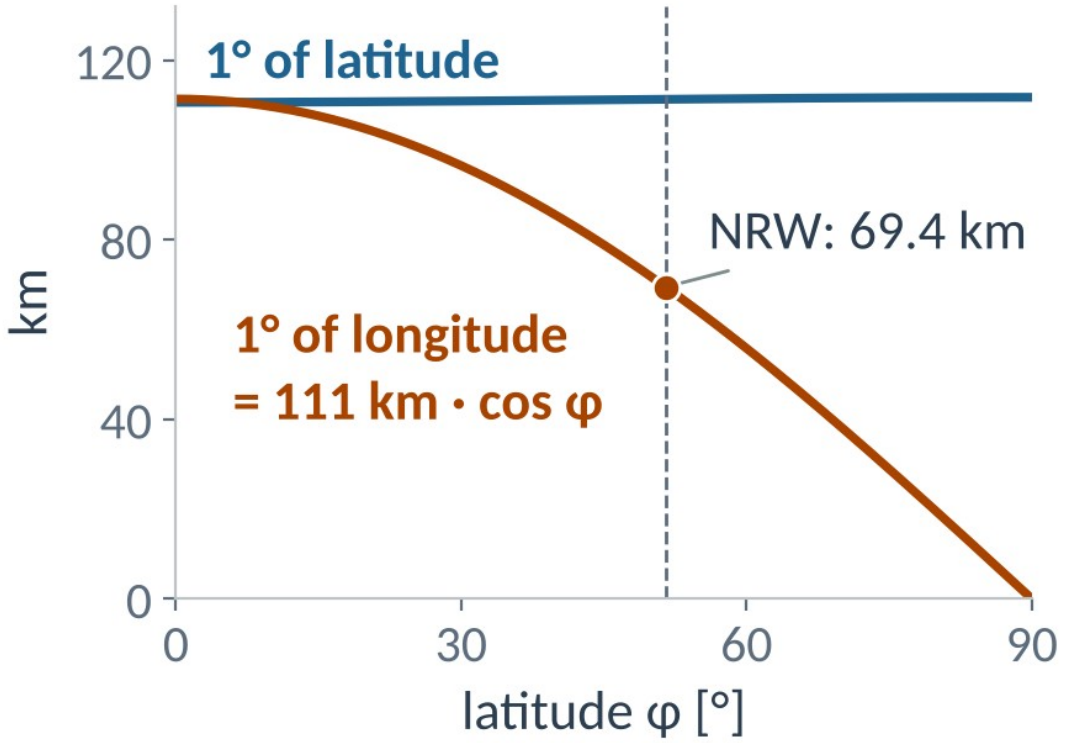
notation	latitude	longitude
degrees, minutes, seconds (DMS)	51° 30′ 00″ N	6° 32′ 00″ E
degrees, decimal minutes	51° 30.000′ N	6° 32.000′ E
decimal degrees	51.500000°	+6.533333°
Boston, signed (west < 0)	42.3601°	−71.0589°

```
lat = 51 + 30/60 + 0/3600  # DMS -> decimal degrees: 51.5
deg = int(lat)             # whole degrees: 51
mins = (lat - deg) * 60    # decimal minutes: 30.000
f"{lat:.6f}"             # 6 decimals (≈ 0.1 m): '51.500000'
```

Wikipedia: Decimal degrees · NMEA 0183



How long is a degree?



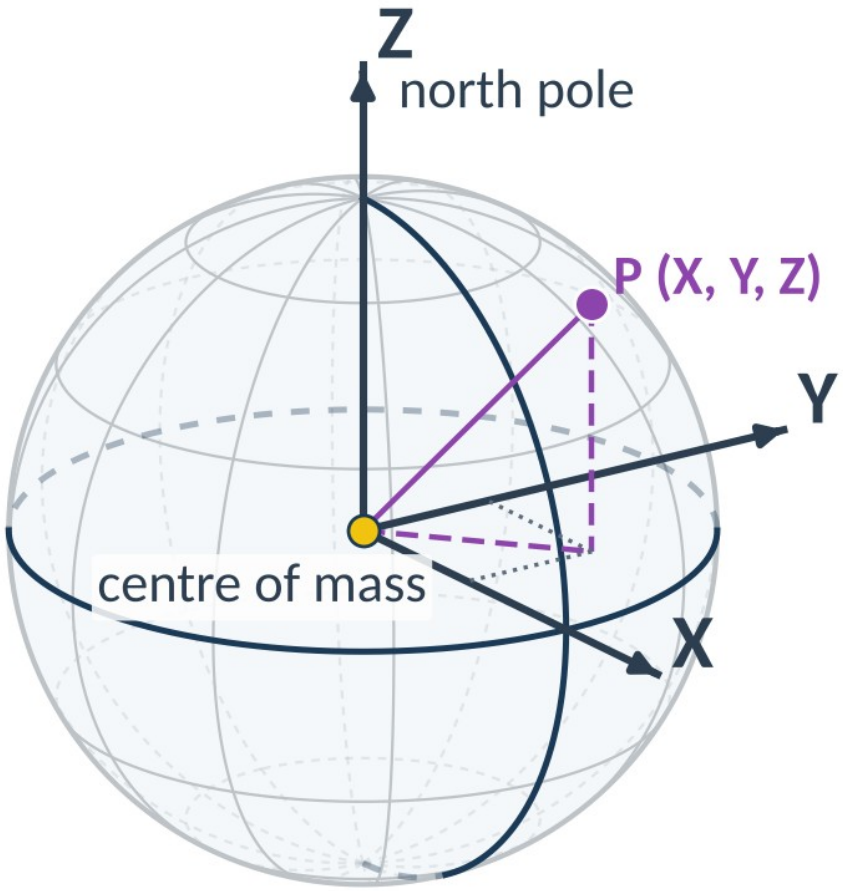
NRW (φ ≈ 51.5°)	north–south	east–west
1°	≈ 111 km	≈ 69 km
1′	≈ 1.85 km	≈ 1.16 km
1″	≈ 31 m	≈ 19 m
0.00001°	≈ 1.1 m	≈ 0.7 m

1′ N–S ≈ 1 nautical mile · NRW: 2.2° ≈ 245 km N–S

```
g = Geod(ellps="GRS80")
g.inv(6, 51.5, 7, 51.5)[2]  # 69 440 m
g.inv(7, 51, 7, 52)[2]      # 111 258 m
```

Wikipedia: Nautical mile · Geographic coordinate system

Geocentric coordinates X, Y, Z



- Origin: centre of mass, Z: rotation axis
- X: (φ, λ) = (0°, 0°) = (0° N, 0° E)
- Y: (φ, λ) = (0°, +90°) = (0° N, 90° E)
- Cartesian, metres, no projection

Kamp-Lintfort (ETRS89)	value
X	3 952 854 m
Y	452 700 m
Z	4 968 417 m

```
t = Transformer.from_crs(
        4937, 4936, always_xy=True)
t.transform(6.5333, 51.5, 70)
```

Wikipedia: Earth-centered, Earth-fixed coordinate system · Helmert transformation



Kamp-Lintfort in four notations

notation	value
DMS	51° 30′ 00″ N 6° 32′ 00″ E
decimal (φ, λ)	(51.500000°, +6.533333°)
(X, Y, Z) [m]	(3 952 854, 452 700, 4 968 417)
UTM 32N (E, N) [m]	(328 794, 5 708 314)

- Same place, different numbers: always state the CRS.
- Compare the UTM values of the two cities.

Pause: which city?

40° 42′ 46″ N
74° 00′ 22″ W

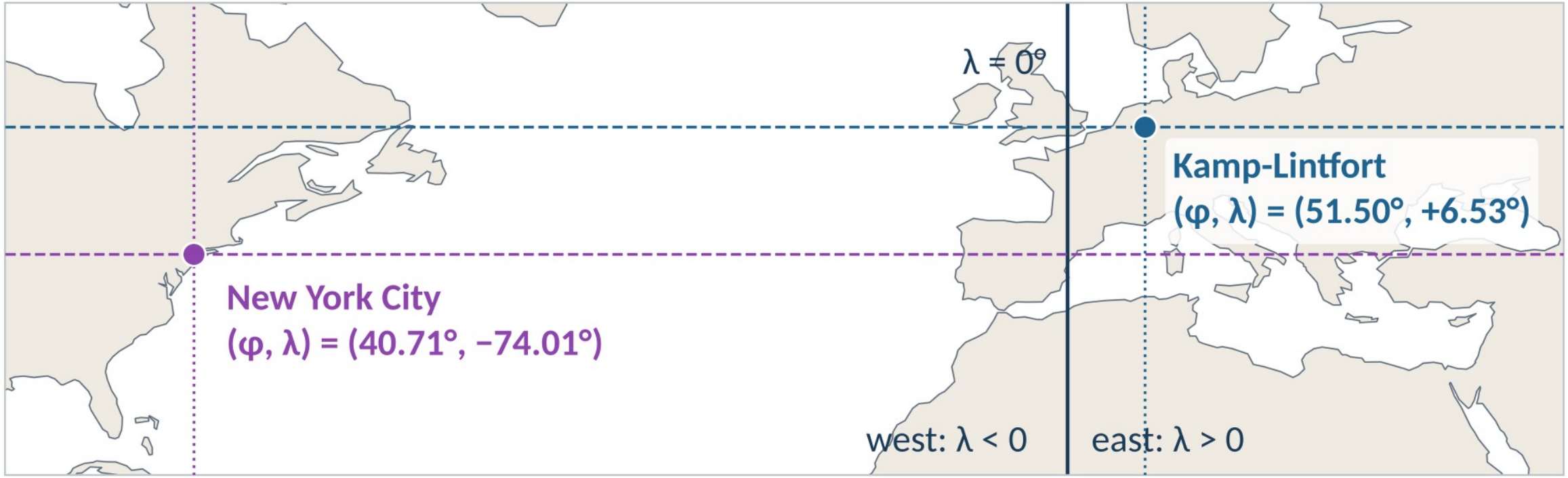
(40.7128°, −74.0060°)

UTM 18N (E, N):
(583 959, 4 507 351)

1 minute with your neighbour



Answer: New York City



11° further south than Kamp-Lintfort. West of Greenwich λ < 0: without the minus sign you land in Kyrgyzstan.

Coastlines and borders: Natural Earth (public domain)

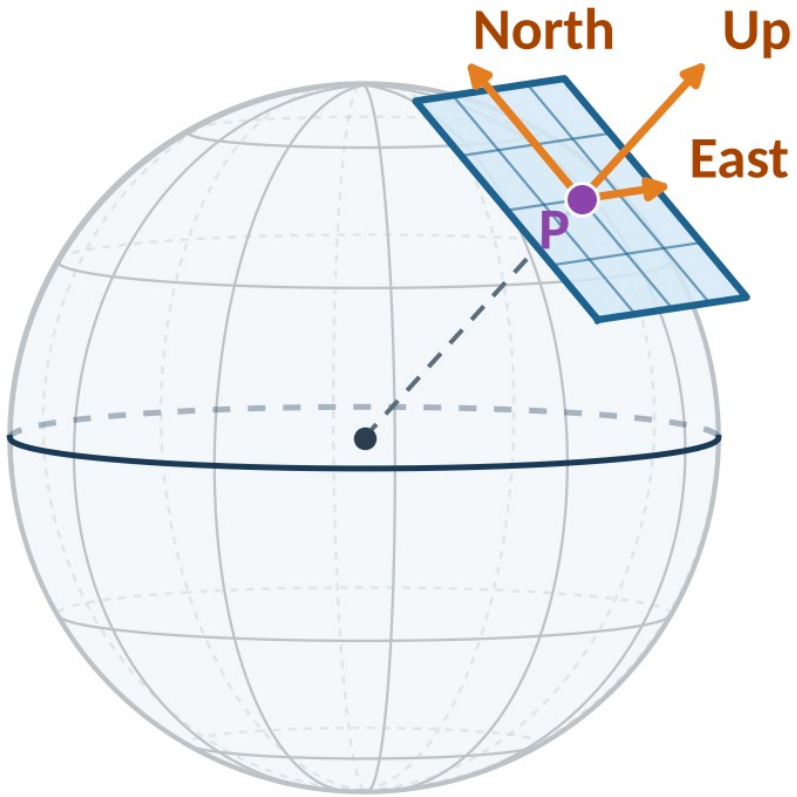
3 Map projections

surfaces · distortion · UTM

Shape of the Earth | Coordinates | **Map projections** | Germany / NRW | CRS in practice



Locally flat: the tangent plane

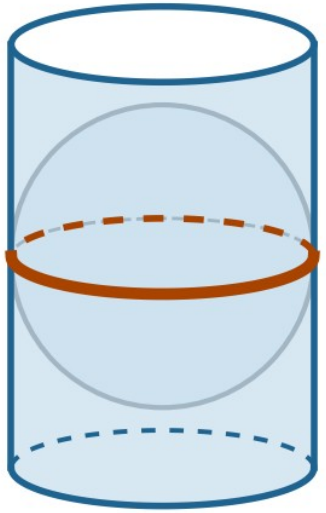


- Near P the Earth is almost flat: the tangent plane is a local map
- Local coordinates East, North, Up in metres
- Lengths fine to ≈ 10 km,
  heights not: curvature

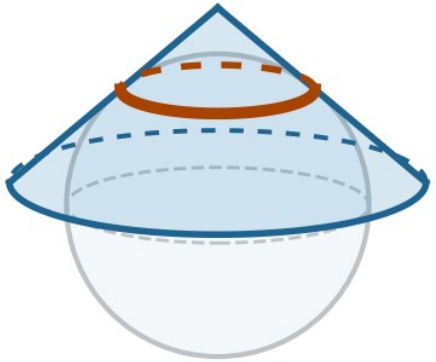
distance from P	plane above sphere	length error
1 km	8 cm	< 1 mm
10 km	7.8 m	8 mm
100 km	785 m	8.2 m

Wikipedia: Local tangent plane coordinates

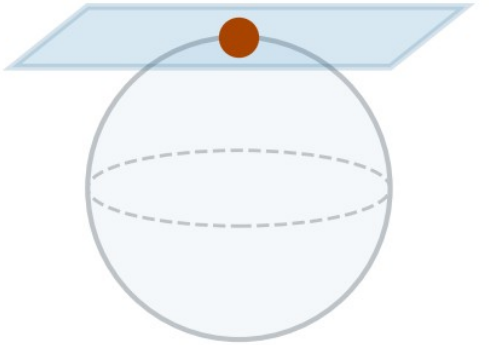
From the ellipsoid to the plane



cylindrical
touches along the equator



conical
touches along a parallel



azimuthal (plane)
touches at one point

- A projection is a formula (φ, λ) → (x, y); a sphere cannot be flattened without distortion.
- No distortion along the line of contact; the surface can be normal, transverse or oblique.

Wikipedia: Map projection · List of map projections

What can a projection preserve?

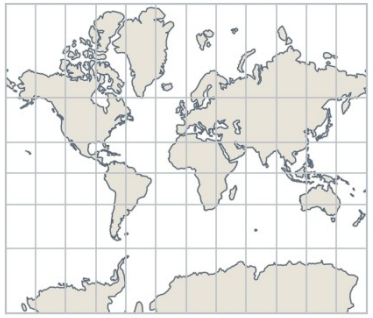
projection class	invariant: what is preserved	where it holds	examples (EPSG)
conformal (orthomorphic)	angles, shapes of small areas	everywhere	Mercator (3395), UTM (25832), stereographic
equal-area (equivalent, authalic)	areas	everywhere	LAEA (3035), Equal Earth (8857), Gall–Peters
equidistant	distances	from one or two points, or along certain lines	azimuthal equidistant, Plate Carrée
azimuthal (zenithal)	directions (azimuths)	only from the centre	stereographic, LAEA, gnomonic
gnomonic	great circles as straight lines	whole map (< hemisphere)	gnomonic

No projection is both conformal and equal-area. Azimuthal combines with the others.

Wikipedia: Map projection · Theorema Egregium



Six projections, six properties



Mercator
conformal: angles



Lambert az. equal-area
equal-area

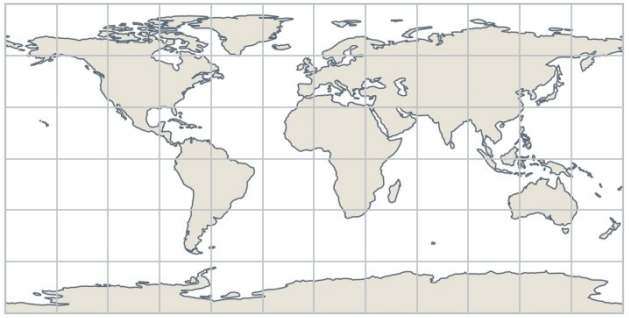


Plate Carrée
equidistant along meridians



Azimuthal equidistant
true distance from centre



Gnomonic
great circles = straight lines



Orthographic
view from space

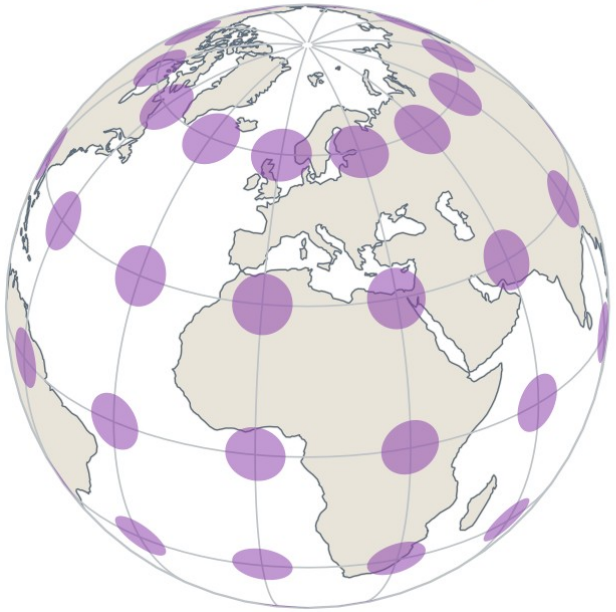
Each projection keeps one property: choose the projection by purpose.

Wikipedia: List of map projections · Gnomonic projection · Lambert azimuthal equal-area projection

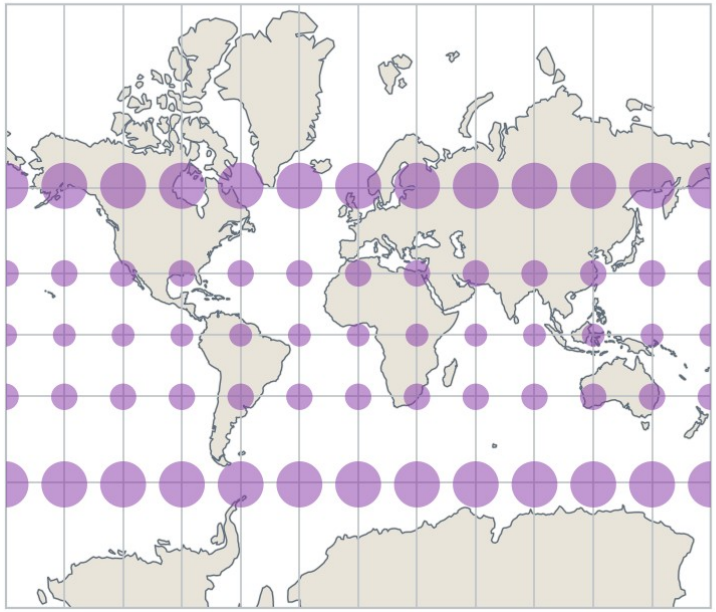
Coastlines and borders: Natural Earth (public domain)

Tissot's indicatrix: one circle, many places

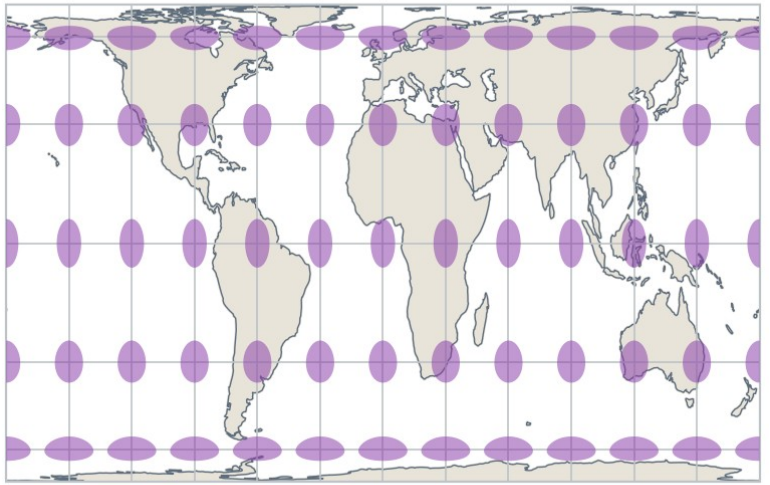
Globe (reference)



Mercator: conformal



Gall–Peters: equal-area



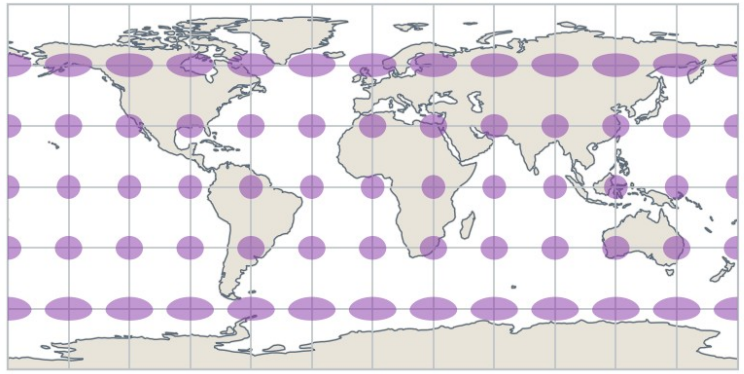
- On the globe all circles are equal (rim flattening is perspective).
- Same cylinder, opposite property: Mercator keeps shapes, Gall–Peters keeps areas.

Wikipedia: Tissot's indicatrix · Mercator projection · Gall–Peters projection

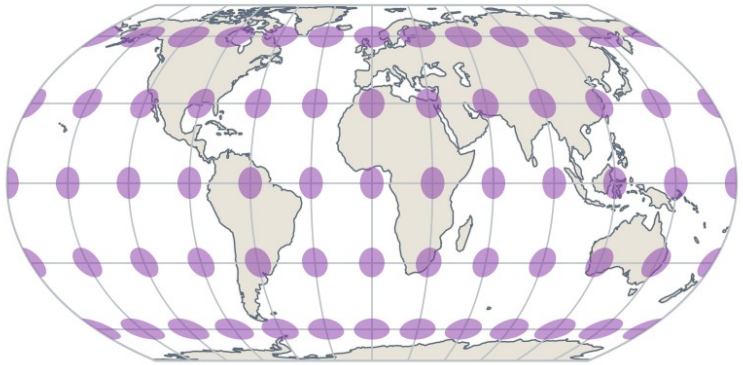
Coastlines and borders: Natural Earth (public domain)

Tissot's indicatrix: more projections

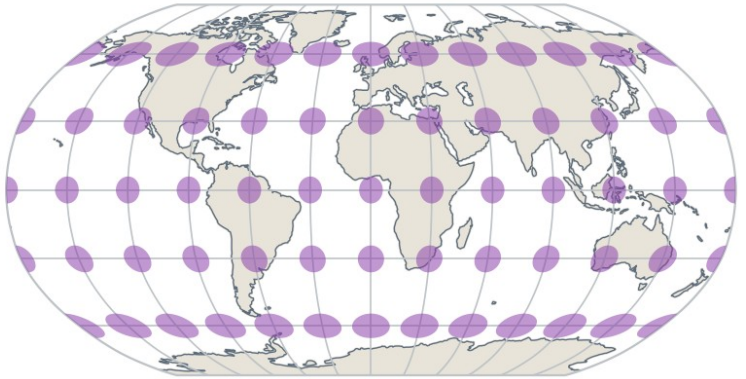
Plate Carrée



Equal Earth: equal-area



Robinson: compromise

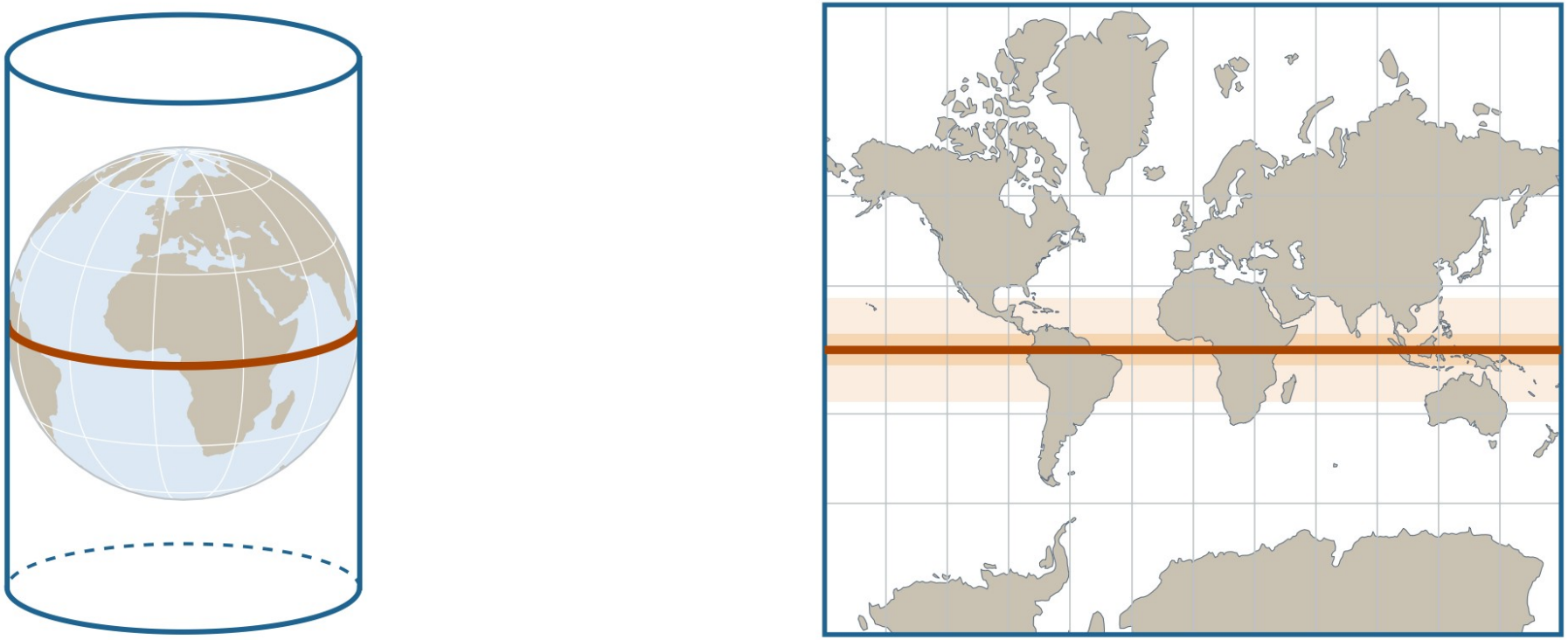


- Plate Carrée = (φ, λ) plotted directly: stretched east–west.
- Equal Earth: equal-area, natural shapes. Robinson: compromise.

Wikipedia: Tissot's indicatrix · Equal Earth projection · Robinson projection

Coastlines and borders: Natural Earth (public domain)

Mercator: the cylinder touches the equator

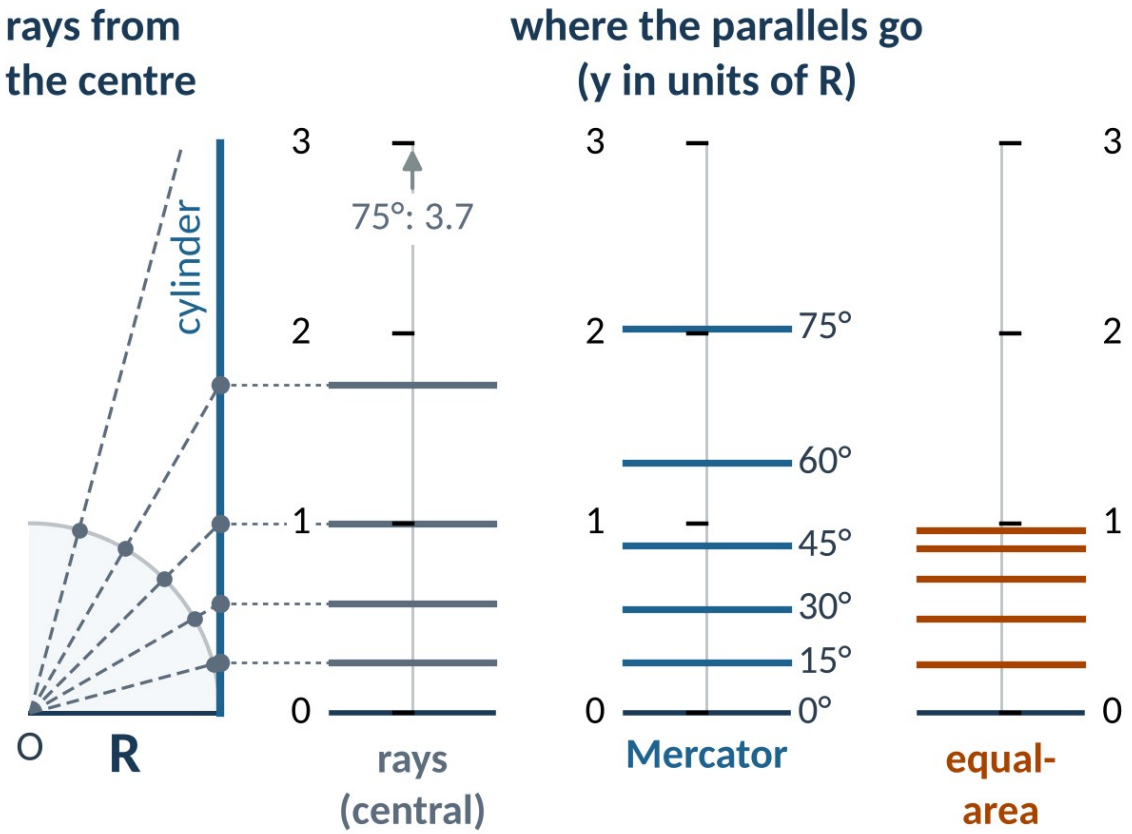


Exact along the equator, increasingly stretched towards the poles.

Wikipedia: Mercator projection

Coastlines and borders: Natural Earth (public domain)

How the Mercator map is built



- Rays from the centre give a different map
- Mercator: east–west stretch 1/cos φ
- Keep shapes ⇒ same stretch north–south: $\Delta y = R\,\Delta\varphi / \cos\varphi$

```
d = np.radians(1/60)  # 1' step
# midpoints of all 1' strips to 75°
phi = np.arange(d/2,
                np.radians(75), d)
# Wright: add up Δy = Δφ / cos φ
y = np.sum(d/np.cos(phi))  # 2.0276
# formula: y = ln tan(45° + φ/2)
np.log(np.tan(np.pi/4 +
       np.radians(75)/2))  # 2.0276
```

Wikipedia: Mercator projection · Central cylindrical projection



Mercator: true angles, wrong areas



million km²	Greenland	Africa
true area	2.2	29.9

- Africa is 14 × larger than Greenland
- On Mercator, Greenland looks even larger than Africa
- Scale grows with 1 / cos φ:
  ×2 at 60°

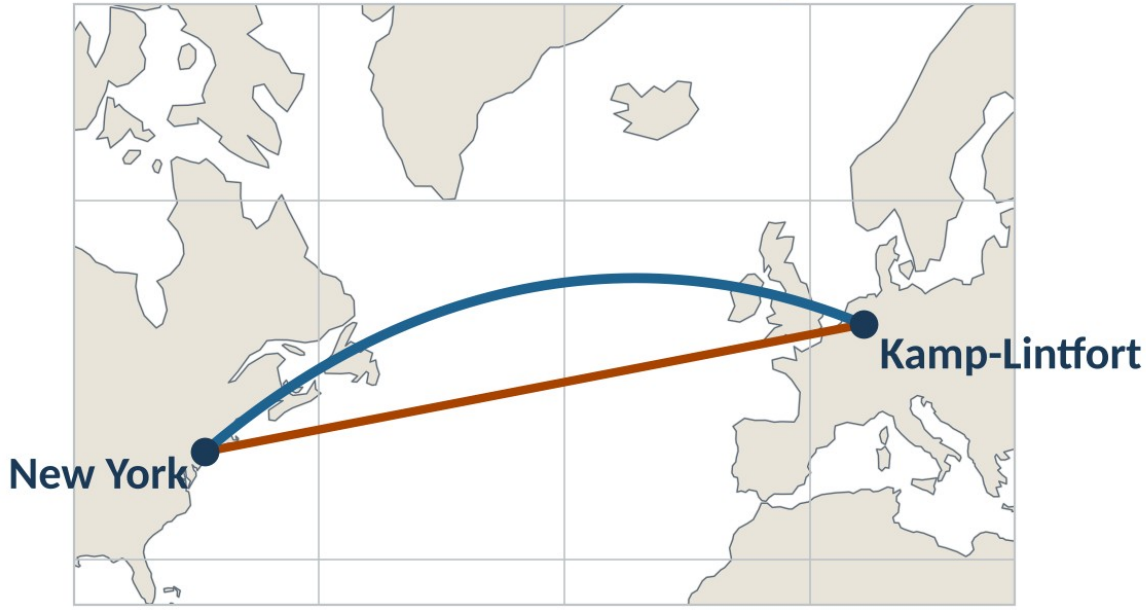
Try it: thetruesize.com

Wikipedia: Mercator projection · Web Mercator projection

Coastlines and borders: Natural Earth (public domain)

Rhumb line or great circle?

Mercator: rhumb line straight



Globe: the great circle goes all around the Earth



rhumb line (loxodrome): constant heading 259° 6 312 km

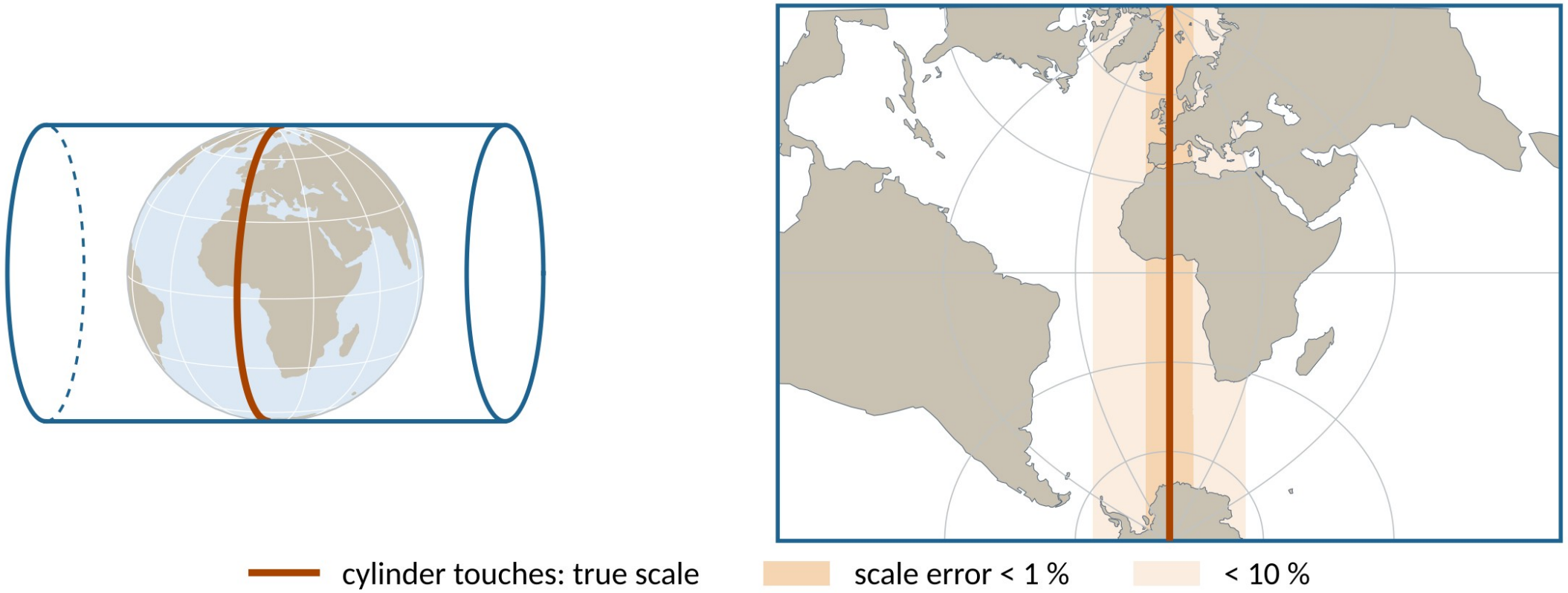
great circle (geodesic): shortest 6 019 km

Shortest is not straight on the map: the great circle saves 294 km.

Wikipedia: Rhumb line · Great circle · Great-circle navigation

Coastlines and borders: Natural Earth (public domain)

Transverse Mercator: the cylinder turned sideways

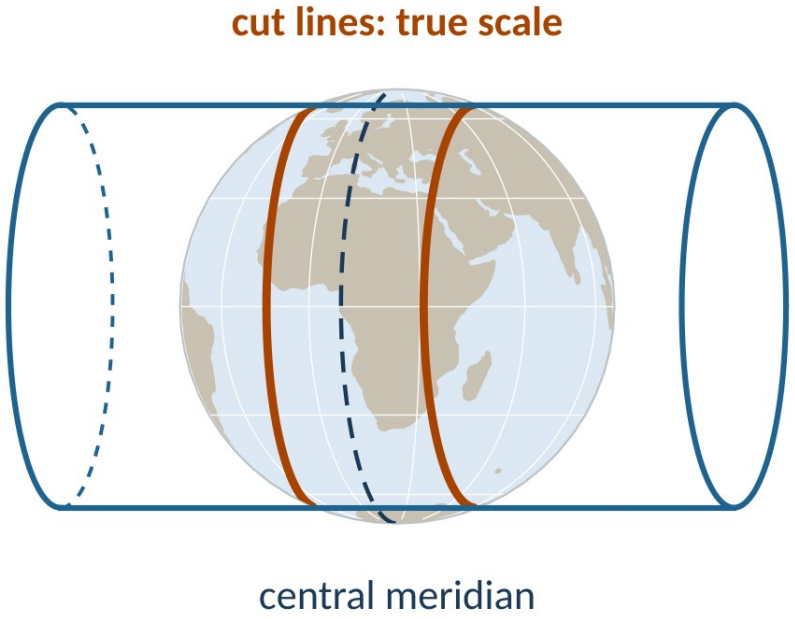


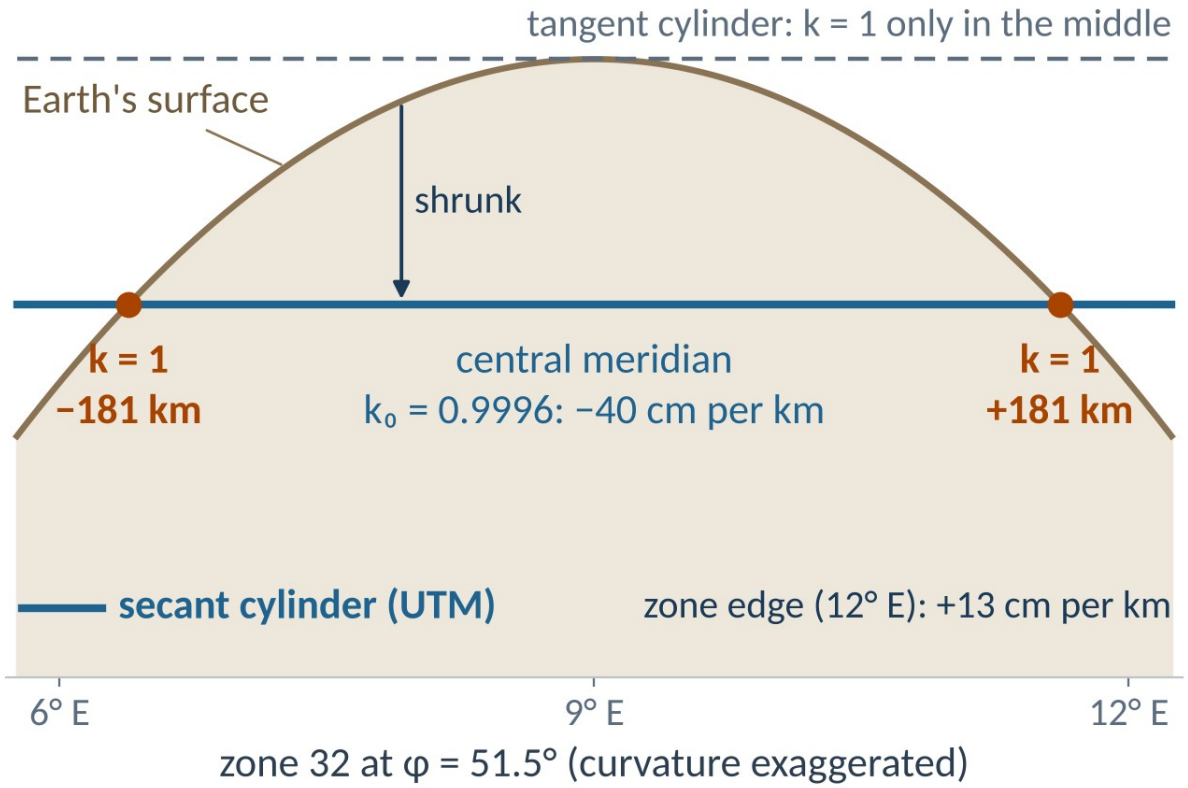
Exact along the central meridian: used in narrow strips (Gauss-Krüger 3°, UTM 6°).

Wikipedia: Transverse Mercator projection

Coastlines and borders: Natural Earth (public domain)

UTM: the secant cylinder



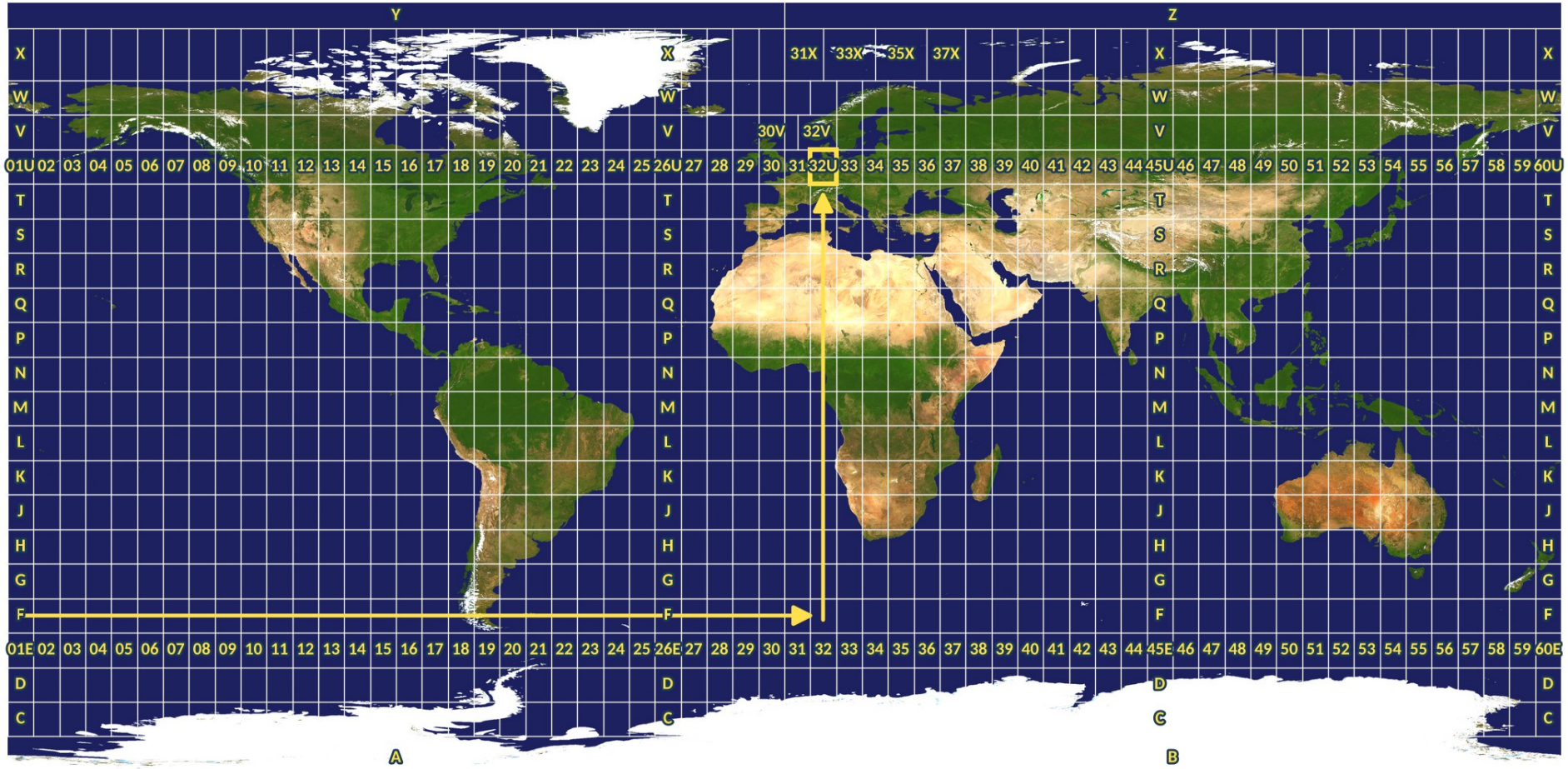


× 0.9996: tangent → secant cylinder, true scale ≈ 181 km east and west

Wikipedia: Universal Transverse Mercator coordinate system

Coastlines and borders: Natural Earth (public domain)

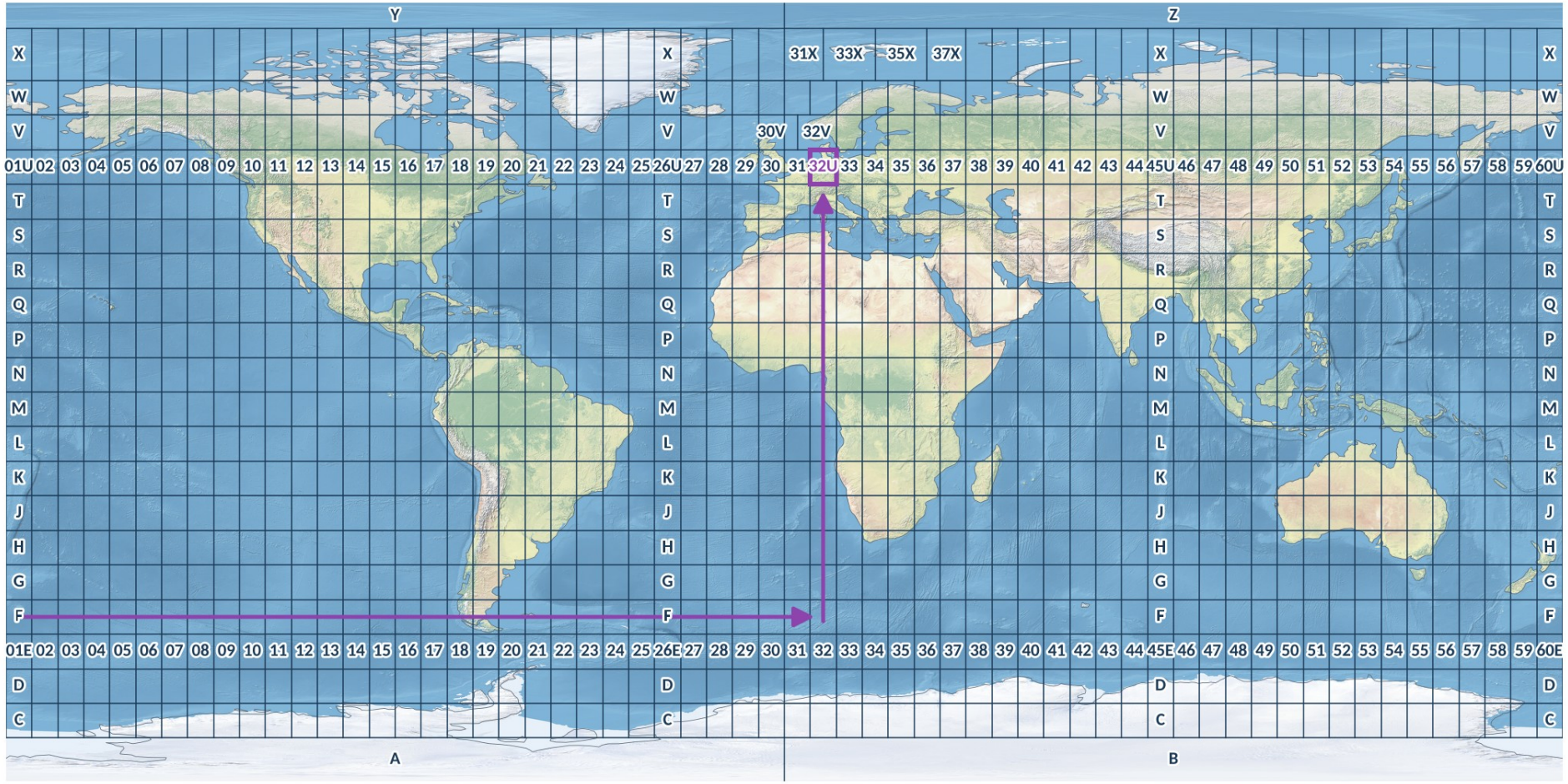
UTM: 60 zones around the world



Wikipedia: Universal Polar Stereographic · Military Grid Reference System

Background: NASA Blue Marble (public domain). Grid: own work.

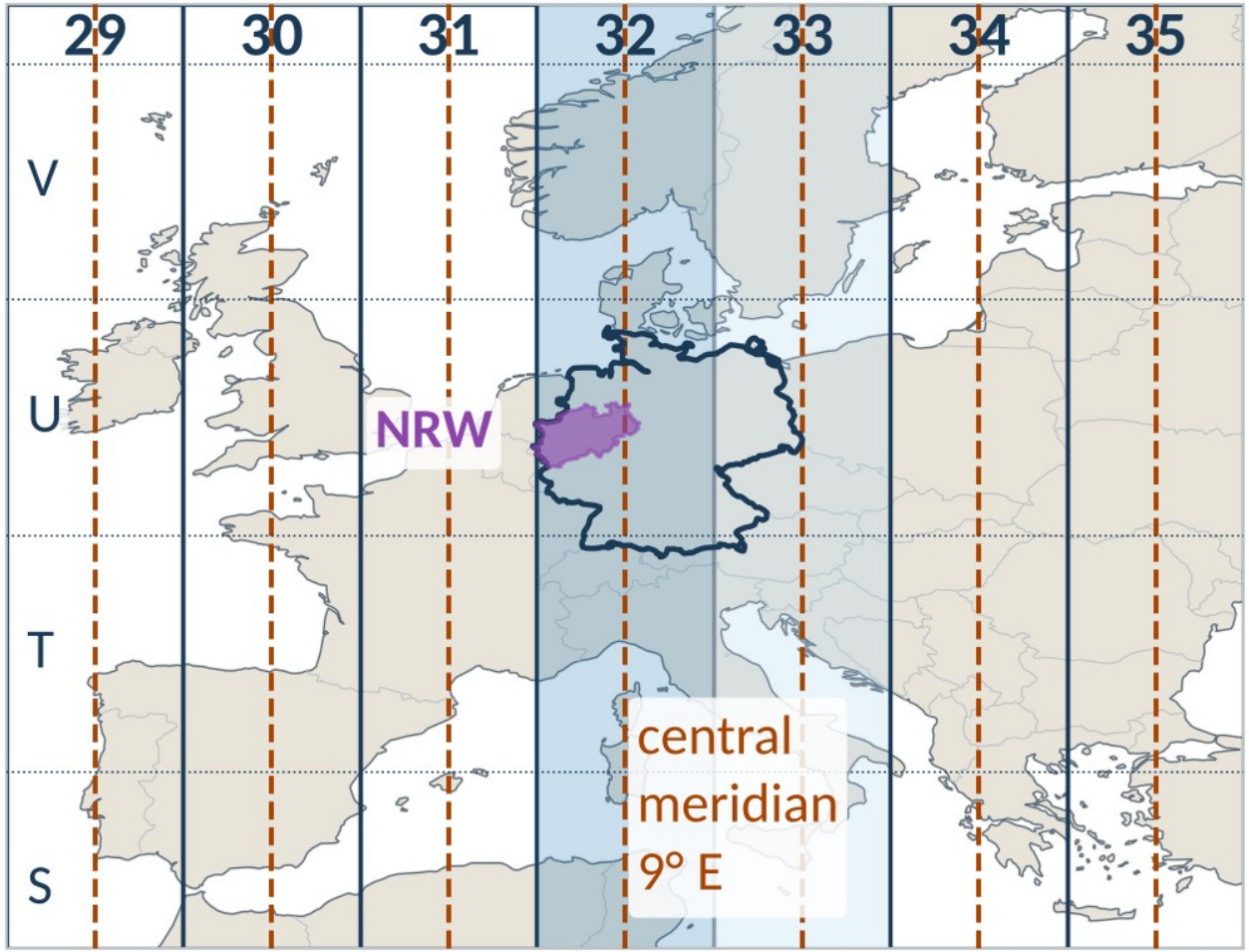
UTM: 60 zones around the world



Wikipedia: Universal Polar Stereographic · Military Grid Reference System

Background: Natural Earth shaded relief (public domain). Grid: own work.

UTM in Europe: zone 32



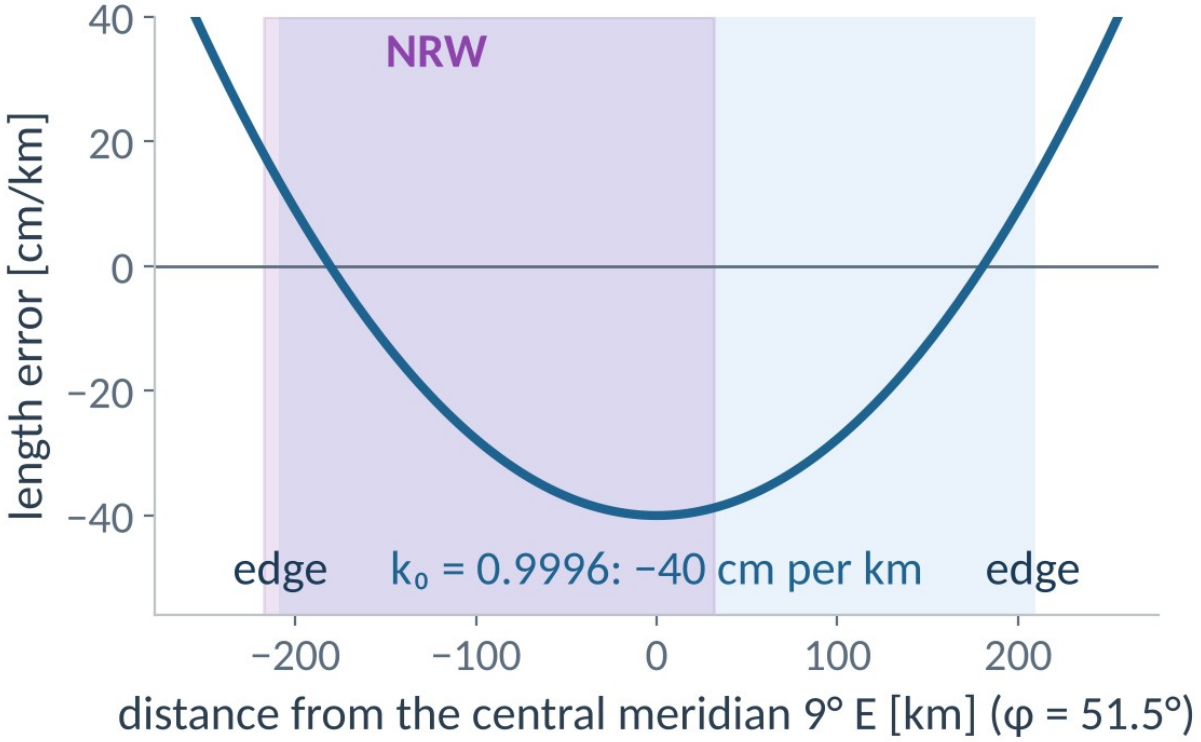
zone 32N	value
longitudes	6° … 12° E
central meridian	9° E
scale factor k_0	0.9996
false easting	500 000 m
datum (Germany)	ETRS89
EPSG	25832

Kamp-Lintfort: zone 32, band U
(E, N) = (328 794 m, 5 708 314 m)

Wikipedia: Universal Transverse Mercator coordinate system

Coastlines and borders: Natural Earth (public domain)

How exact is UTM in NRW?



- Central meridian: −40 cm per km
- True scale ≈ 181 km east and west
- Zone edge (φ = 51.5°): +13 cm per km

```
p = Proj("EPSG:25832")  # UTM 32N
# scale factor k at Kamp-Lintfort
f = p.get_factors(6.5333, 51.5)
f.meridional_scale  # 0.99996
```

Wikipedia: Universal Transverse Mercator coordinate system



Which projection for what?

purpose	projection	property	EPSG
maps and analysis in NRW / Germany	UTM zone 32N	conformal	25832
statistics and areas in Europe	Lambert azimuthal equal-area	equal-area	3035
world thematic maps	Equal Earth	equal-area	8857
web maps (display only)	Web Mercator	conformal (sphere)	3857
navigation charts	Mercator	conformal	3395
storing and exchanging data	none: (φ, λ)	–	4326 / 4258

Wikipedia: EPSG Geodetic Parameter Dataset · European Terrestrial Reference System 1989



4 Germany / NRW

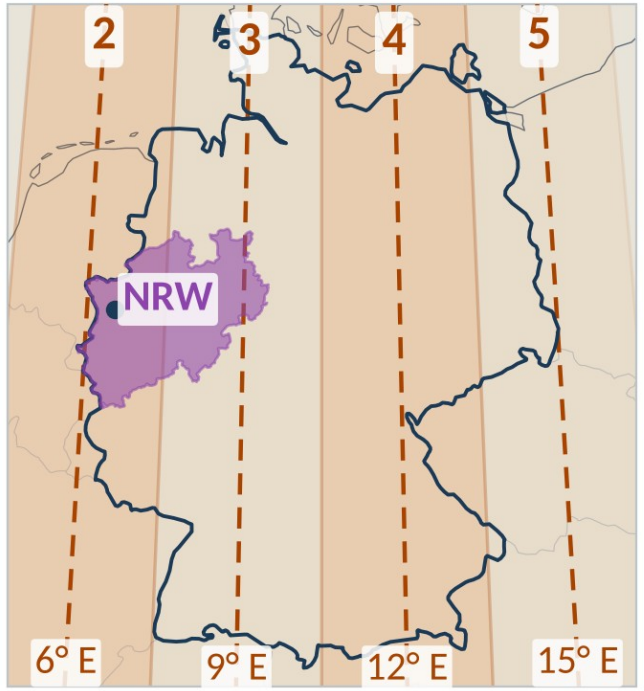
DHDN · Gauss-Krüger · ETRS89 · DHHN2016

Shape of the Earth | Coordinates | Map projections | **Germany / NRW** | CRS in practice

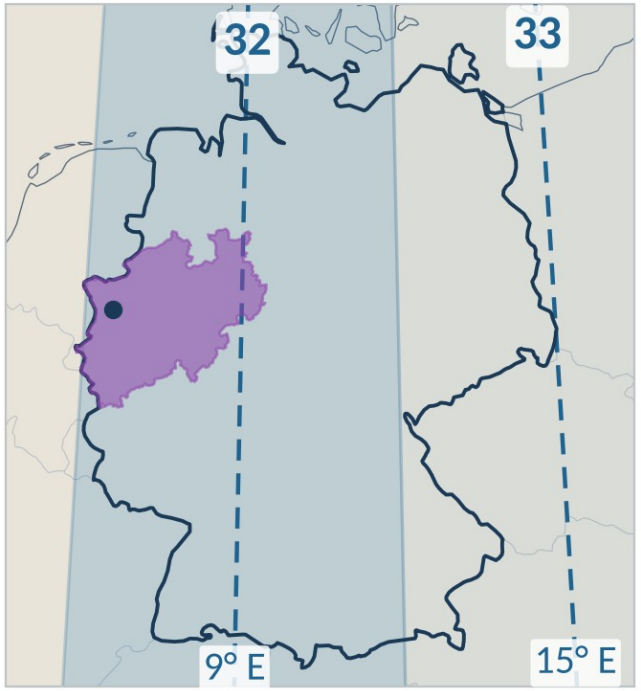


Old and new: DHDN / Gauss-Krüger → ETRS89 / UTM

Gauss-Krüger: 3° zones



UTM: 6° zones



	DHDN / GK	ETRS89 / UTM
ellipsoid	Bessel 1841	GRS80
datum	regional	geocentric
zone width	3°	6°
CM in NRW	6° E, 9° E	9° E
k_0	1	0.9996
E (Kamp-L.)	2 537 078	328 794
EPSG	31466/31467	25832/4647

2 537 078: zone 2,
37 km east of 6° E

Wikipedia: Gauss–Krüger coordinate system · ETRS89

Coastlines and borders: Natural Earth (public domain)

EPSG: a unique ID for every CRS

- **EPSG**: European Petroleum Survey Group, registry today run by IOGP
- A unique numeric key for every registered CRS
- **EPSG:4326** WGS 84, geographic (GPS)
- **EPSG:25832** ETRS89 / UTM zone 32N
  - Kamp-Lintfort: 328 794, 5 708 314
- **EPSG:4647** ETRS89 / UTM zone 32N (zE-N)
  - Kamp-Lintfort: 32 328 794, 5 708 314 (zone prefix 32)
- **EPSG:3857** WGS 84 / Pseudo-Mercator: OpenStreetMap, Google, Bing

Wikipedia: EPSG Geodetic Parameter Dataset



Kamp-Lintfort in eight CRSs

CRS	EPSG	east value	north value	unit
WGS 84, geographic	4326	+6.533333°	51.500000°	degree
ETRS89, geographic	4258	+6.533333°	51.500000°	degree
DHDN, geographic	4314	+6.534036°	51.501317°	degree
DHDN / Gauss-Krüger zone 2	31466	2 537 078	5 707 408	m
DHDN / Gauss-Krüger zone 3	31467	3 328 800	5 710 156	m
ETRS89 / UTM zone 32N	25832	328 794	5 708 314	m
ETRS89 / UTM 32N (zE-N)	4647	32 328 794	5 708 314	m
WGS 84 / Pseudo-Mercator	3857	727 287	6 710 219	m (web)

```
for epsg in (4314, 31466, 31467, 25832, 4647):  # target CRSs
    t = Transformer.from_crs(4258, epsg, always_xy=True)  # from ETRS89
    print(epsg, t.transform(6.5333, 51.5))  # (east, north)
```

Wikipedia: Spatial reference system



Recognise a CRS from its numbers

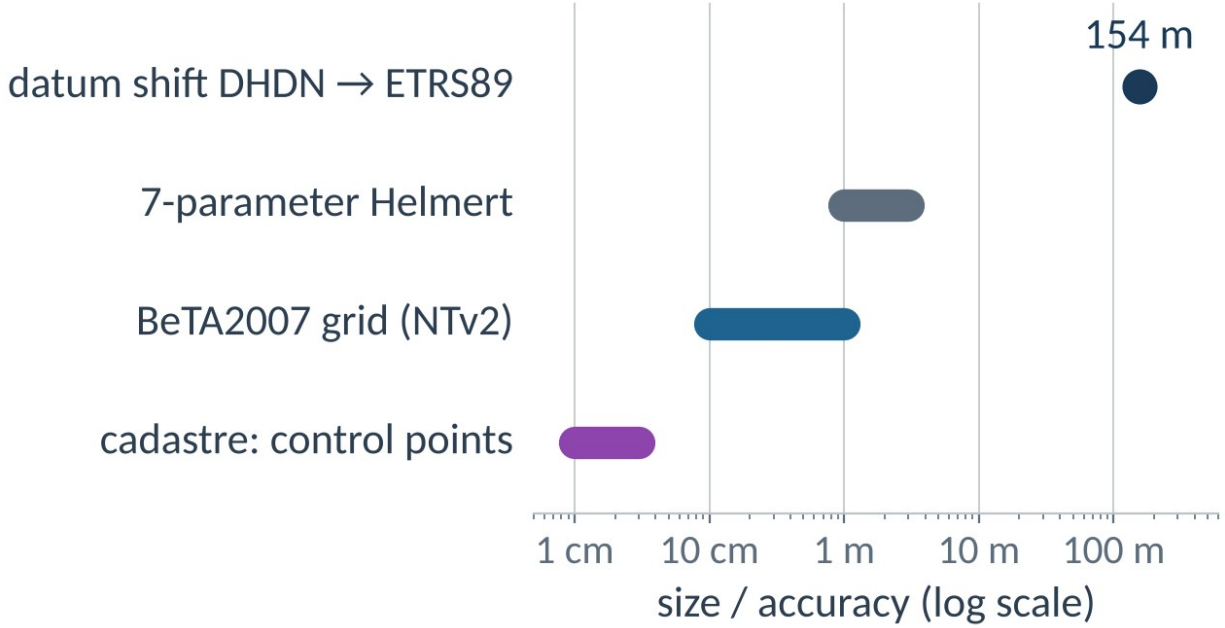
pattern of the east value	east range	north range	probably
≈ 6 … 9.5, with decimals	5.9 … 9.5	50.3 … 52.5	degrees: 4258/4326
7 digits, starts with 2	2 490 000 … 2 739 000	5 576 000 … 5 825 000	GK zone 2: 31466
7 digits, starts with 3	3 279 000 … 3 532 000	5 580 000 … 5 821 000	GK zone 3: 31467
6 digits	279 000 … 532 000	5 578 000 … 5 819 000	UTM 32N: 25832
8 digits, starts with 32	32 279 000 … 32 532 000	5 578 000 … 5 819 000	UTM 32N: 4647
6–7 digits, north ≈ 6.6 M	651 000 … 1 053 000	6 502 000 … 6 895 000	Web Mercator: 3857

Count the digits, read the first ones. In degrees, check which value is the latitude.

Wikipedia: EPSG Geodetic Parameter Dataset



DHDN → ETRS89 is a transformation



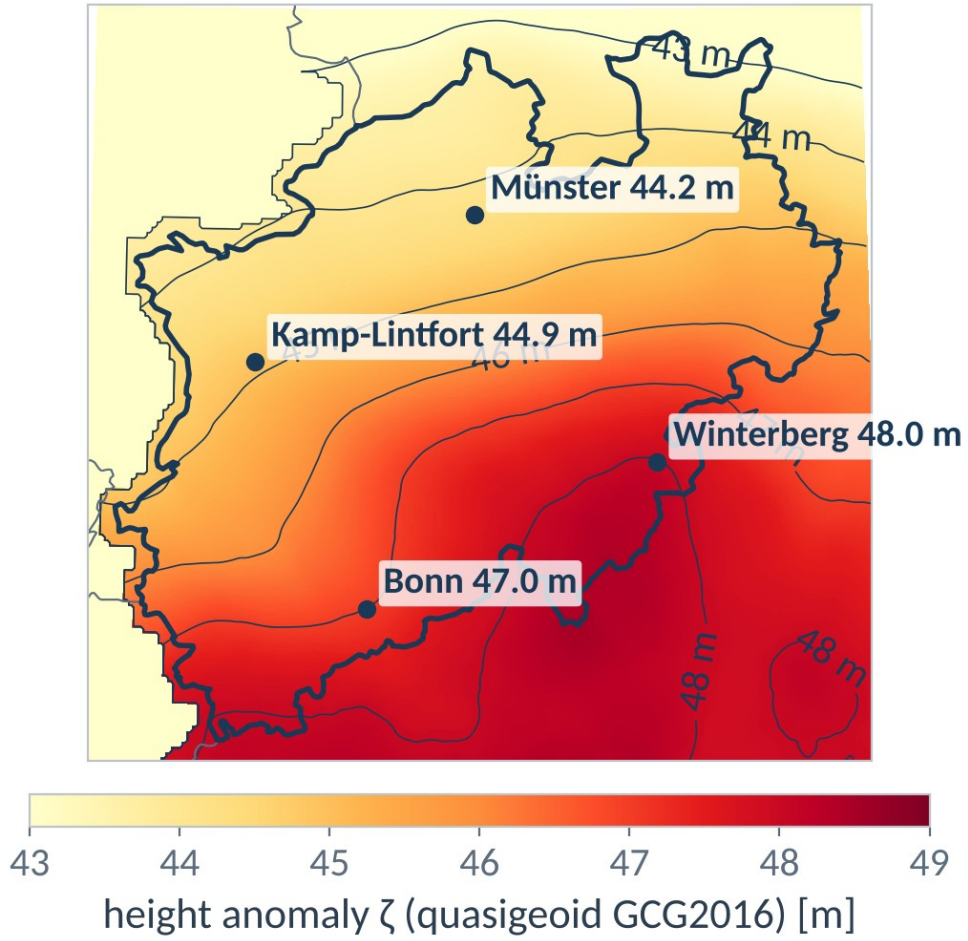
- New datum: new ellipsoid and new position
- Same numbers ≈ 154 m apart
- The method decides the accuracy

```
tg = TransformerGroup(4314, 4258, always_xy=True)  # DHDN -> ETRS89
[(t.description, t.accuracy) for t in tg.transformers]  # best first
# Kamp-Lintfort: BeTA2007 grid vs Helmert = 1.8 m
```

Wikipedia: Helmert transformation · Geodetic datum



Heights in Germany: DHHN2016

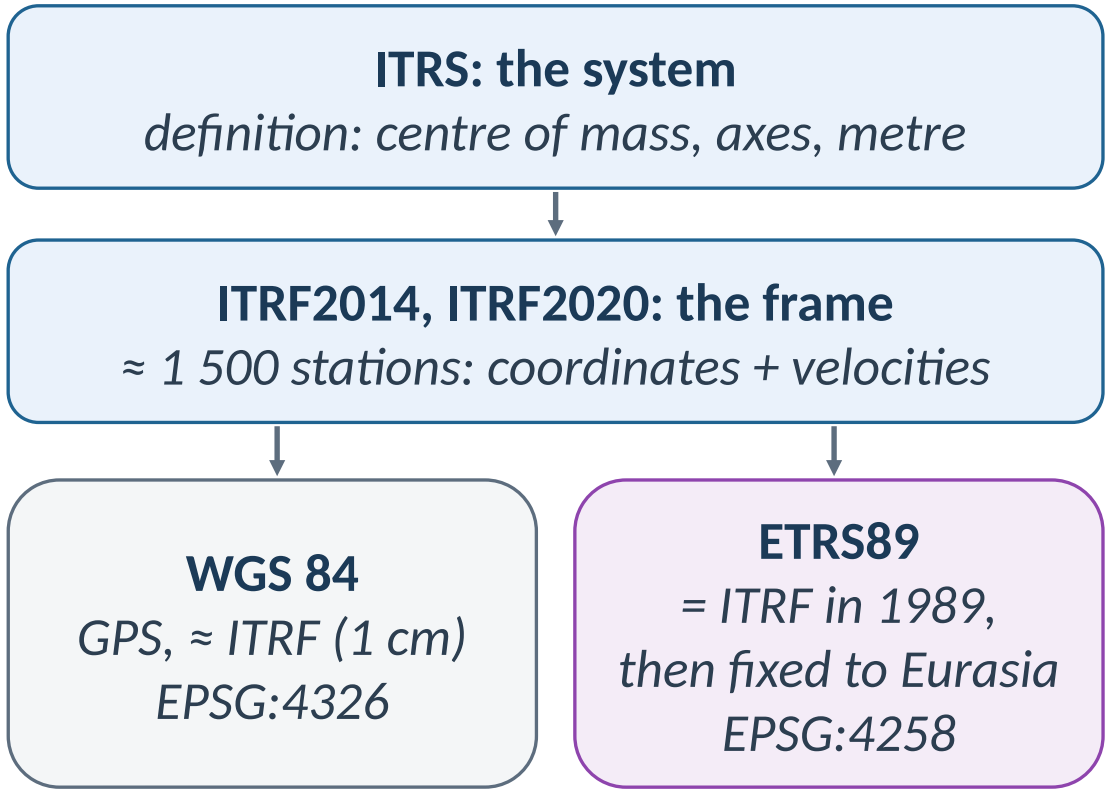


- **h** GNSS, ellipsoidal (ETRS89)
- **H** maps, DEM NRW: DHHN2016 (NHN)
- **H = h − ζ**, ζ from GCG2016 (< 1 cm)
- NRW: ζ ≈ 44 … 48 m

```
t = Transformer.from_crs(4937, 9924,
    always_xy=True)  # uses GCG2016
t.transform(6.5333, 51.5, 100.0)
# h = 100 m -> H = 55.13 m
```

Wikipedia: Normal height · Geoid

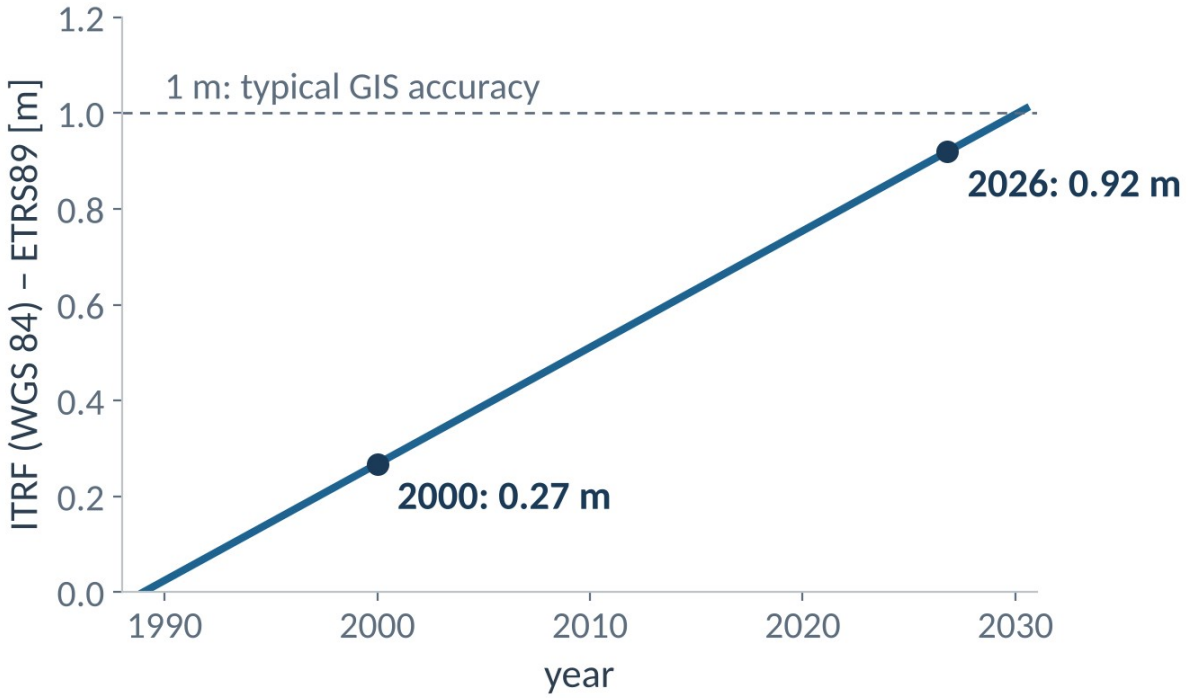
ITRF: the global reference frame



- The plates move: in ITRF, Germany moves ≈ 2.5 cm per year north-east
- ITRF / WGS 84: grid fixed to the whole Earth, coordinates change
- ETRS89: grid glued onto Europe, coordinates stay
- Kamp-Lintfort 2026: 0.92 m apart

Wikipedia: International Terrestrial Reference System and Frame

ETRS89 is not WGS 84



- ETRS89 is fixed to the Eurasian plate (1989)
- WGS 84 is global: Europe moves ≈ 2.4 cm per year north-east
- Kamp-Lintfort 2026: 0.92 m apart
- GIS at 1 m: treat as equal
  GNSS / RTK: use ETRS89 (SAPOS)

Wikipedia: ETRS89 · Plate tectonics

Sources and further reading (1/2)

From the Earth to a map: Spatial reference system · Map projection

WGS 84: the ellipsoid with a, b and f: World Geodetic System · Flattening

Reference ellipsoids: Earth ellipsoid · GRS 80 · World Geodetic System · Flattening

The geoid: where is 'zero height'? Geoid · Earth Gravitational Model · Orthometric height

Datum: where does the ellipsoid sit? Geodetic datum · Figure of the Earth

Geographic coordinates: latitude φ and longitude λ: Geographic coordinate system · Latitude · Longitude · Prime meridian

Writing degrees: Decimal degrees · NMEA 0183

How long is a degree? Nautical mile · Geographic coordinate system

Geocentric coordinates X, Y, Z: Earth-centered, Earth-fixed coordinate system · Helmert transformation

Locally flat: the tangent plane: Local tangent plane coordinates

From the ellipsoid to the plane: Map projection · List of map projections

What can a projection preserve? Map projection · Theorema Egregium

Six projections, six properties: List of map projections · Gnomonic projection · Lambert azimuthal equal-area projection

Tissot's indicatrix: one circle, many places: Tissot's indicatrix · Mercator projection · Gall–Peters projection

Tissot's indicatrix: more projections: Tissot's indicatrix · Equal Earth projection · Robinson projection

Mercator: the cylinder touches the equator: Mercator projection

How the Mercator map is built: Mercator projection · Central cylindrical projection

Mercator: true angles, wrong areas: Mercator projection · Web Mercator projection

Data: Natural Earth, EGM96 (public domain); BeTA2007, GCG2016 (open licence) via PROJ-data. Figures: own work (make_slides.py), CC BY-SA 4.0.



Sources and further reading (2/2)

Rhumb line or great circle? Rhumb line · Great circle · Great-circle navigation

Transverse Mercator: the cylinder turned sideways: Transverse Mercator projection

UTM: the secant cylinder: Universal Transverse Mercator coordinate system

UTM: 60 zones around the world: Universal Polar Stereographic · Military Grid Reference System

UTM in Europe: zone 32: Universal Transverse Mercator coordinate system

How exact is UTM in NRW? Universal Transverse Mercator coordinate system

Which projection for what? EPSG Geodetic Parameter Dataset · European Terrestrial Reference System 1989

Old and new: DHDN / Gauss-Krüger → ETRS89 / UTM: Gauss–Krüger coordinate system · ETRS89

EPSG: a unique ID for every CRS: EPSG Geodetic Parameter Dataset

Kamp-Lintfort in eight CRSs: Spatial reference system

Recognise a CRS from its numbers: EPSG Geodetic Parameter Dataset

DHDN → ETRS89 is a transformation: Helmert transformation · Geodetic datum

Heights in Germany: DHHN2016: Normal height · Geoid

ITRF: the global reference frame: International Terrestrial Reference System and Frame

ETRS89 is not WGS 84: ETRS89 · Plate tectonics

Data: Natural Earth, EGM96 (public domain); BeTA2007, GCG2016 (open licence) via PROJ-data. Figures: own work (make_slides.py), CC BY-SA 4.0.

